

CBSE Class 10 Mathematics — Solutions

1. Two water taps together can fill a tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.

Answer:

Largertap : 15hours, Smallertap : 25hours

Solution:

- Let the time taken by the smaller tap to fill the tank be x hours. Then the larger tap takes $(x - 10)$ hours.
- Part of tank filled by smaller tap in 1 hour = $\frac{1}{x}$. Part of tank filled by larger tap in 1 hour = $\frac{1}{x-10}$.
- Together they fill the tank in $9\frac{3}{8} = \frac{75}{8}$ hours. So, part filled together in 1 hour = $\frac{1}{\frac{75}{8}} = \frac{8}{75}$.
- Equation: $\frac{1}{x} + \frac{1}{x-10} = \frac{8}{75}$.
- Multiply both sides by $75x(x - 10)$: $75(x - 10) + 75x = 8x(x - 10)$.
- Simplify: $75x - 750 + 75x = 8x^2 - 80x \Rightarrow 150x - 750 = 8x^2 - 80x$.
- Bring all terms to one side: $0 = 8x^2 - 80x - 150x + 750 \Rightarrow 8x^2 - 230x + 750 = 0$.
- Divide by 2: $4x^2 - 115x + 375 = 0$. Solve by quadratic formula:
 $x = \frac{115 \pm \sqrt{115^2 - 4 \cdot 4 \cdot 375}}{2 \cdot 4} = \frac{115 \pm \sqrt{13225 - 6000}}{8} = \frac{115 \pm \sqrt{7225}}{8} = \frac{115 \pm 85}{8}$.
- Thus $x = \frac{115+85}{8} = \frac{200}{8} = 25$ or $x = \frac{115-85}{8} = \frac{30}{8} = 3.75$.
- If $x = 3.75$, then $x - 10 = -6.25$, which is not possible (time cannot be negative). Hence $x = 25$ hours for smaller tap and $x - 10 = 15$ hours for larger tap.

Derivation: Let x = time for smaller tap. $\frac{1}{x} + \frac{1}{x-10} = \frac{8}{75}$
 $75(x - 10) + 75x = 8x(x - 10)$
 $150x - 750 = 8x^2 - 80x$
 $8x^2 - 230x + 750 = 0$
 $4x^2 - 115x + 375 = 0$
 $x = \frac{115 \pm \sqrt{13225 - 6000}}{8} = \frac{115 \pm 85}{8}$
 $x = 25$ or $x = 3.75$ (rejected) Thus smaller tap = 25 hours, larger tap = 15 hours. *Quadratic Equations — Word Problems (Time and Work)*

2. If the roots of the quadratic equation $(a - b)x^2 + (b - c)x + (c - a) = 0$ are equal, prove that $2a = b + c$.

Answer:

Proved : $2a = b + c$

Solution:

- The given equation is $(a - b)x^2 + (b - c)x + (c - a) = 0$.
- For equal roots, the discriminant $D = 0$.
- Here, $A = (a - b)$, $B = (b - c)$, $C = (c - a)$.
- Compute discriminant: $D = B^2 - 4AC = (b - c)^2 - 4(a - b)(c - a)$.

5. Expand $(b - c)^2 = b^2 - 2bc + c^2$.
6. Expand $4(a-b)(c-a) = 4[(a-b)(c-a)] = 4[ac - a^2 - bc + ab] = 4ac - 4a^2 - 4bc + 4ab$.
7. Set $D = 0$: $b^2 - 2bc + c^2 - (4ac - 4a^2 - 4bc + 4ab) = 0$.
8. Simplify: $b^2 - 2bc + c^2 - 4ac + 4a^2 + 4bc - 4ab = 0$.
9. Combine like terms: $b^2 + c^2 + 4a^2 + (-2bc + 4bc) - 4ac - 4ab = b^2 + c^2 + 4a^2 + 2bc - 4ac - 4ab = 0$.
10. Recognise the perfect square: $(b + c)^2 = b^2 + c^2 + 2bc$ and $(2a)^2 = 4a^2$. Also $-4ac - 4ab = -4a(c + b)$.
11. Rewrite as: $(b + c)^2 + 4a^2 - 4a(b + c) = 0 \Rightarrow (b + c)^2 - 4a(b + c) + (2a)^2 = 0$.
12. This is $(b + c - 2a)^2 = 0$, hence $b + c - 2a = 0$, so $2a = b + c$.

Derivation:

$$D = (b-c)^2 - 4(a-b)(c-a) = 0b^2 - 2bc + c^2 - 4(ac - a^2 - bc + ab) = 0b^2 - 2bc + c^2 - 4ac + 4a^2 + 4bc - 4ab = 0$$

Quadratic Equations — Nature of Roots (Equal Roots Proof)